

# Computer Networks

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## Lecture 17: Data Link Layer

# Content

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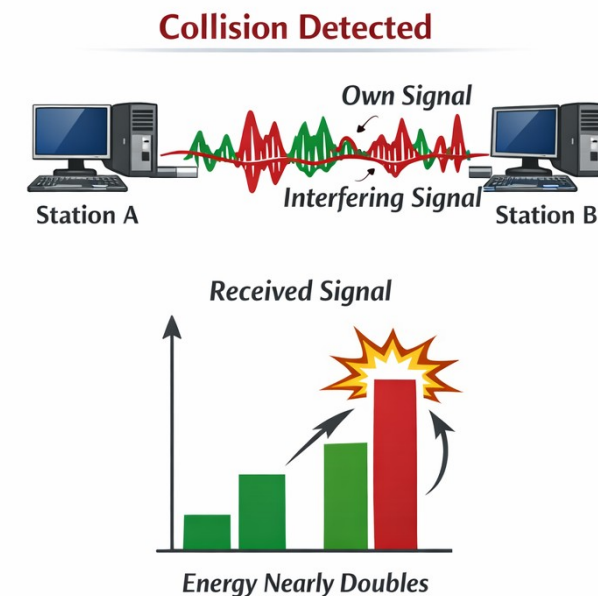
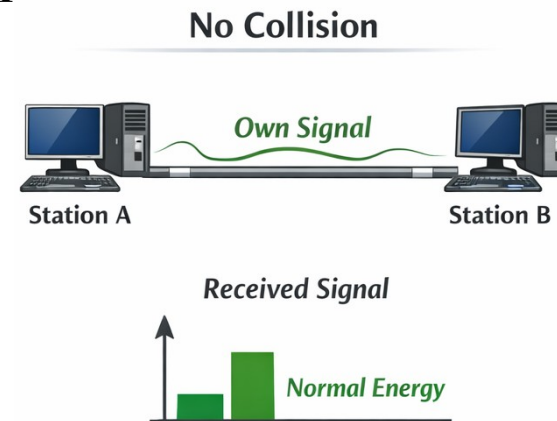


- **CSMA/CA**

# Introduction

In Carrier Sense Multiple Access with Collision Detection, a station must listen to the medium while transmitting to detect whether a collision occurs.

- While transmitting, a station also monitors the signal on the channel.
- **No Collision Case**
  - The station receives only its own transmitted signal.
- **Collision Case**
  - The station receives two signals simultaneously:
    - Its own signal
    - Signal from another transmitting station
- The second signal adds extra energy to the channel.

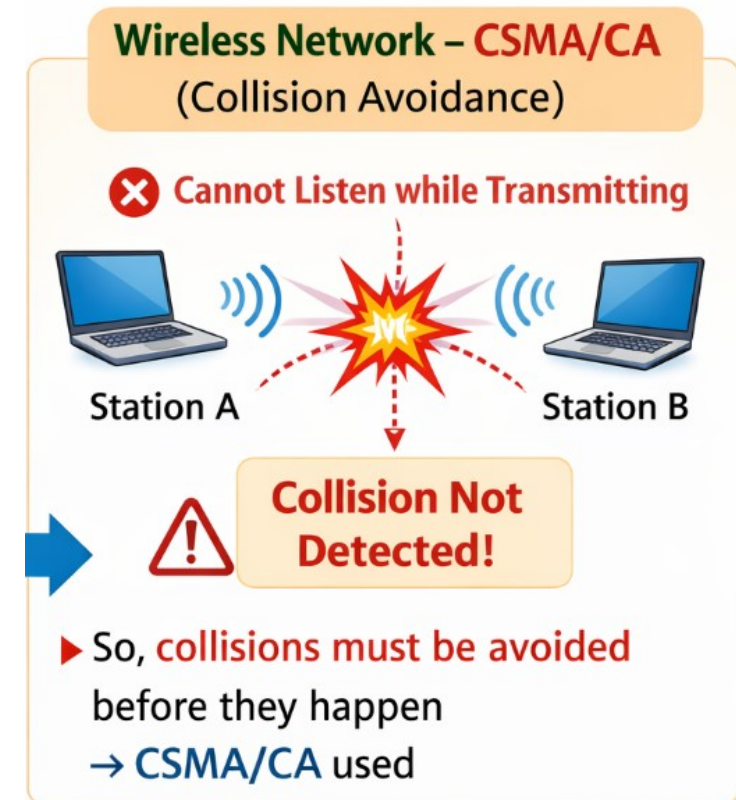


# CSMA/CA

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In a wireless network, much of the sent energy is lost in transmission. The received signal has very little energy. Therefore, a collision may add only 5 to 10 percent additional energy. This is not useful for effective collision detection.

- In wireless networks, a station cannot listen while transmitting, so collisions cannot be detected.
- Therefore, wireless networks use Carrier Sense Multiple Access with Collision Avoidance to prevent collisions before they happen.



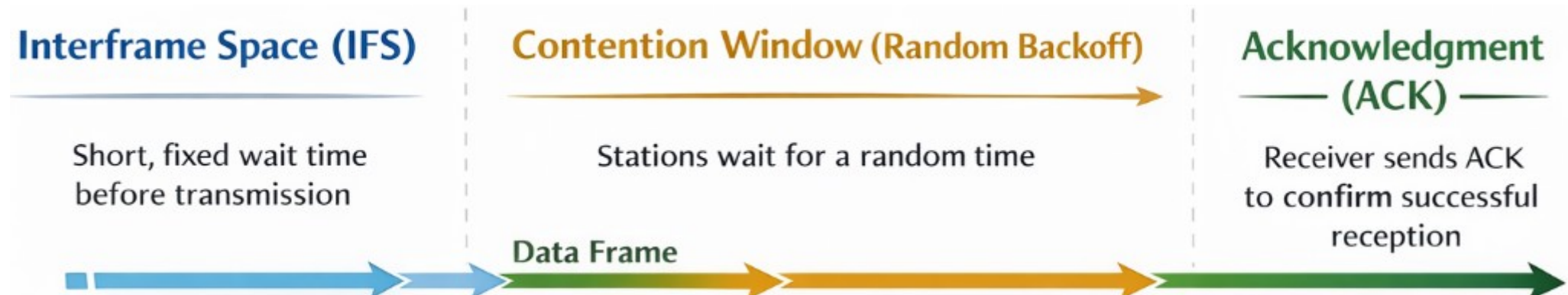
# CSMA/CA

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## Strategies Used in CSMA/CA

Collisions are avoided through the use of CSMA/CA's three strategies:

- The inter frame space
- The contention window
- The acknowledgments



# CSMA/CA

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## Interframe Space

- When a station senses the channel as idle, it does not transmit immediately.
- The station waits for a fixed Interframe Space (IFS) before attempting transmission.
- This delay allows signals from distant stations to reach the station.
- If the channel remains idle after the IFS period, the station proceeds to the contention process before sending data.
- Different IFS durations can be assigned to different stations or frame types.

# CSMA/CA

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## Contention Window

The Contention Window is a time interval divided into small slots. A station that wants to transmit randomly selects a number of slots as its waiting time.

- Initially, the contention window is very small (1 slot).
- If the station cannot transmit (channel busy or collision), the window doubles in size.
- This process continues:  $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16$  slots, etc.

# CSMA/CA

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## Contention Window: Example

Consider That the size of Contention Window is 0–7 slots and a station randomly selects 3 slots.

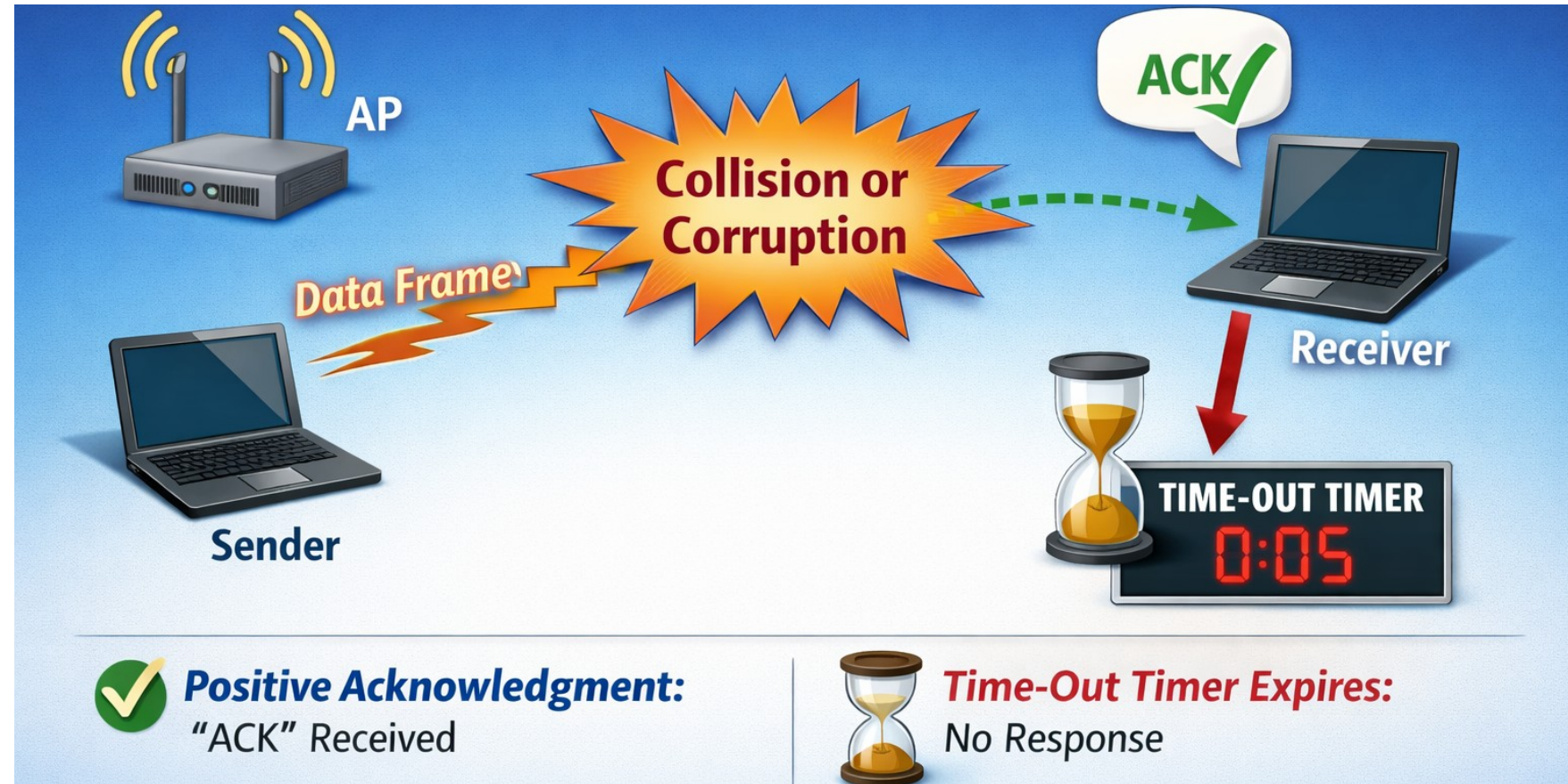
|           |                |                 | <u>Channel Becomes Busy During Countdown</u> |                |                  |
|-----------|----------------|-----------------|----------------------------------------------|----------------|------------------|
| Time Slot | Channel Status | Backoff Counter | Time Slot                                    | Channel Status | Backoff Counter  |
| Slot 1    | Idle           | 3 → 2           | Slot 1                                       | Idle           | 3 → 2            |
| Slot 2    | Idle           | 2 → 1           | Slot 2                                       | Busy           | Timer paused     |
| Slot 3    | Idle           | 1 → 0           | Slot 3                                       | Busy           | Timer paused     |
|           |                |                 | Slot 4                                       | Idle           | 2 → 1            |
|           |                |                 | Slot 5                                       | Idle           | 1 → 0 → Transmit |



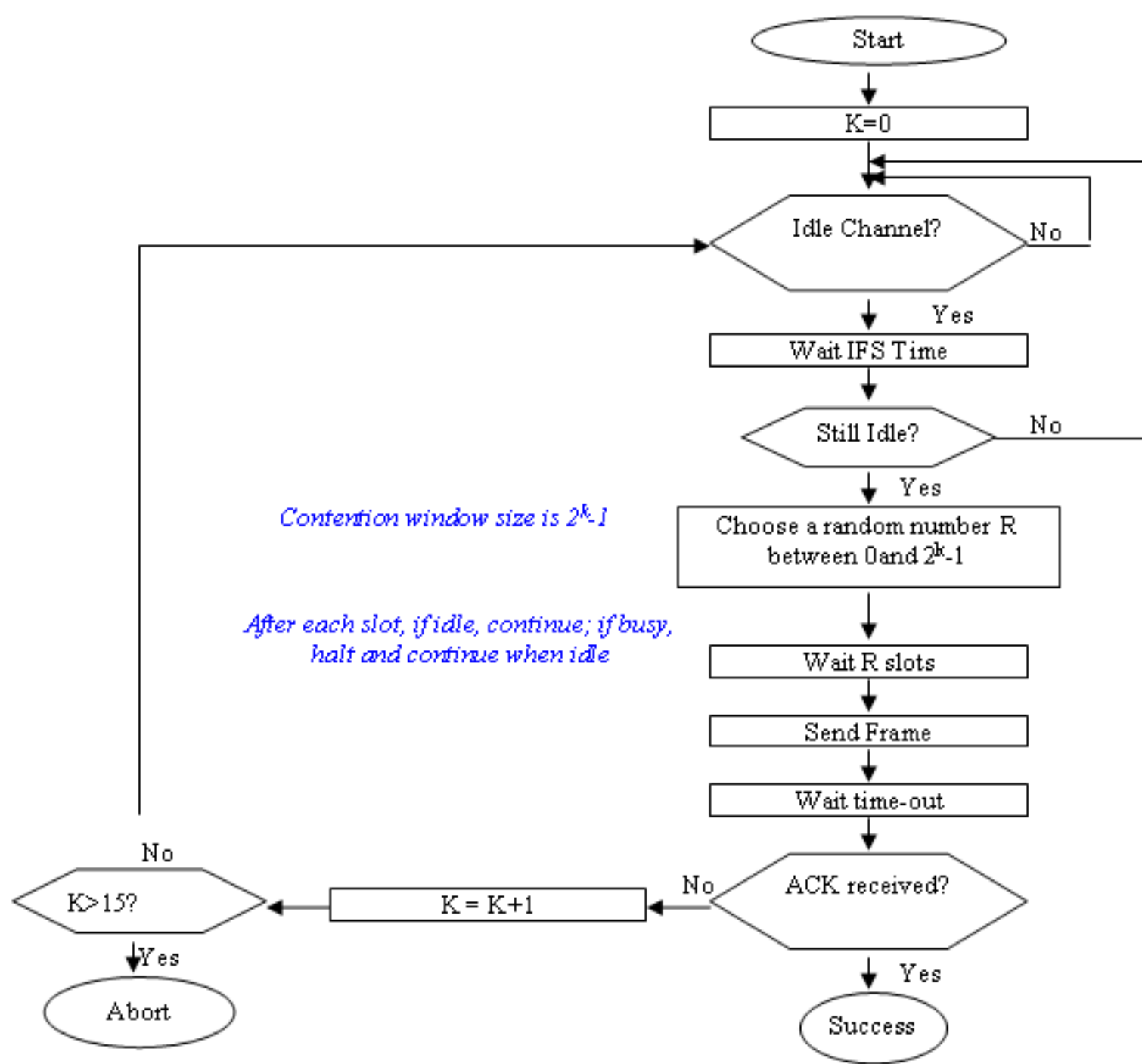
# CSMA/CA

## Acknowledgement

The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.



# CSMA/CA



*Flow diagram for the CSMA/CA*

# CSMA/CA

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## Vulnerable Time

- In wireless networks, signal propagation takes time.
- Two stations may sense the channel idle at the same time.
- Both may start transmitting almost simultaneously.
- Their signals collide

The vulnerable period is approximately equal to the propagation delay ( $T_P$ ).

# Questions

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1. Three stations A, B, and C want to transmit in a wireless network using CSMA/CA. After the IFS, they select the following random backoff slots from a contention window of 0–7.
  - A = 3 slots
  - B = 5 slots
  - C = 2 slotsSlot duration =  $20 \mu s$ . Which station transmits the message first.
2. A station chooses a backoff value = 6 slots. Slot time =  $10 \mu s$ . After 3 slots, another station starts transmitting and keeps the channel busy for  $80 \mu s$ . Find the total time after IFS when the station will transmit.

# Questions

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3. Given that in CSMA/CA the minimum contention window size  $CW_{\min} = 2$  slots. A station experiences 4 transmission failures .Find the new contention window size.
4. Two wireless stations are 600 m apart. Signal propagation speed =  $3 \times 10^8$  m/s. Find the vulnerable time in CSMA/CA.
5. If the propagation delay between two farthest stations in a CSMA/CD network is  $5 \mu\text{s}$ , the maximum time required to detect a collision is